

WEEK 3 - ADVANCED CYBERSECURITY PRACTICAL

Network Security Assessment & Traffic Investigation

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Kali Linux in VMware Workstation • 21 September 2026

Scope: Authorized virtual lab assessment only.

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3. Introduction & Objectives

This practical demonstrates a controlled network-security assessment of an authorized VMware lab using Nmap and Wireshark. The workflow covers discovery, service enumeration, traffic observation, evidence-based risk assessment and hardening verification.

Objectives: identify authorized hosts; identify exposed services; inspect DNS/TCP/UDP/ICMP/ARP traffic; correlate Nmap results with packet behavior; document findings; and verify security improvements.

4. Lab Environment & Network Architecture

The supplied evidence shows Kali Linux running in VMware Workstation with eth0 as the active lab interface. The Nmap evidence targets 192.168.56.0/24. IP/MAC values that were already redacted in the supplied screenshots remain redacted.

Kali Linux (Scanner)	→	VMware Lab Network	→	Authorized Lab Hosts
eth0	→	192.168.56.0/24	→	3 hosts reported up

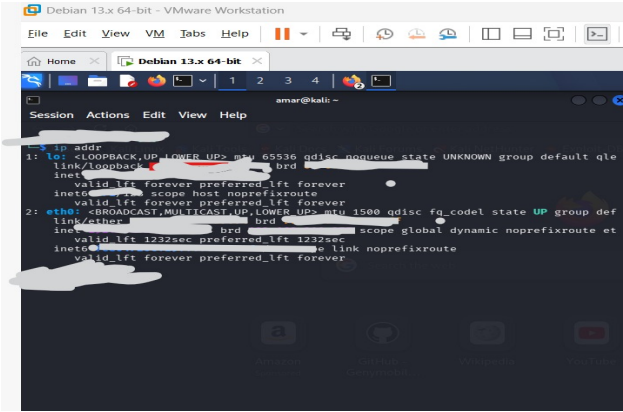


Figure 1. Original Kali `ip addr` evidence

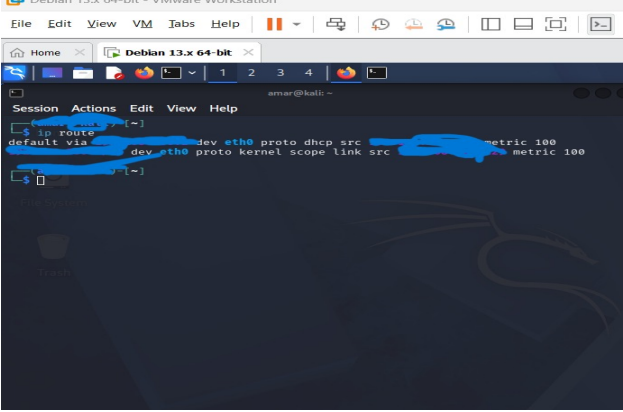


Figure 2. Original Kali `ip route` evidence.

5. Task 7 - Network Discovery

Zenmap/Nmap was used against 192.168.56.0/24. The supplied evidence shows an Nmap 7.99 scan covering 256 addresses. The terminal result reports 3 hosts up.

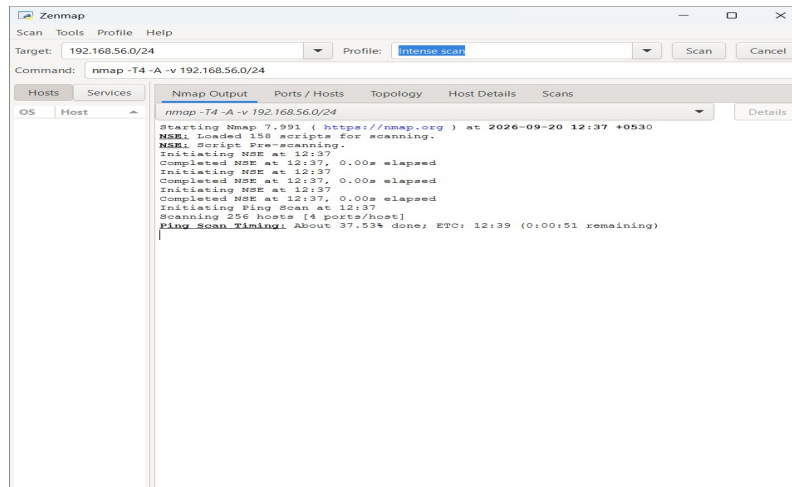


Figure 3. Zenmap scan of the lab range.

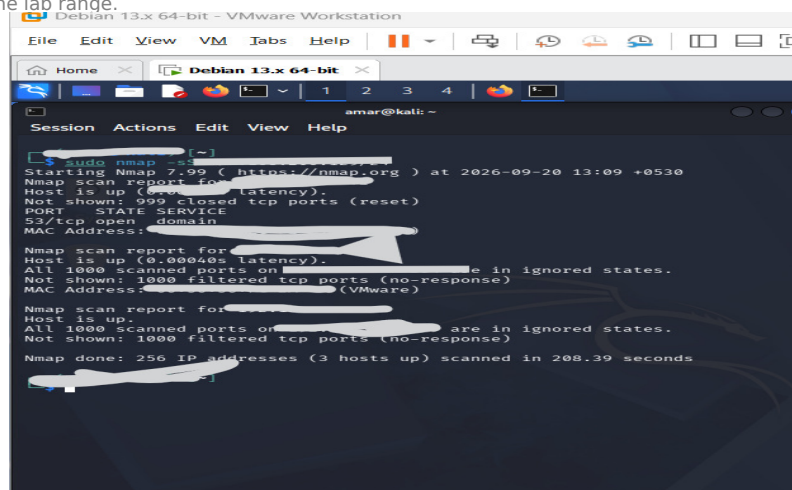


Figure 4. Terminal Nmap summary showing 3 hosts up.

6. Task 8 - Advanced Nmap Assessment

The supplied evidence includes OS detection, service/version detection, a UDP top-port scan and an NSE vulnerability-script scan.

Assessment	Observed evidence	Interpretation
Service/version	53/tcp open; domain/dnsmasq 2.80 visible	DNS service is exposed; validate necessity and scope.
OS detection	VMware/Windows-related fingerprint guesses	Fingerprint is indicative, not definitive.
UDP scan	53/udp visible; other ports filtered/closed	Validate DNS requirement and network restriction.
NSE vuln scan	Scan executed; no confirmed CVE visible	Do not invent a vulnerability; verify manually.

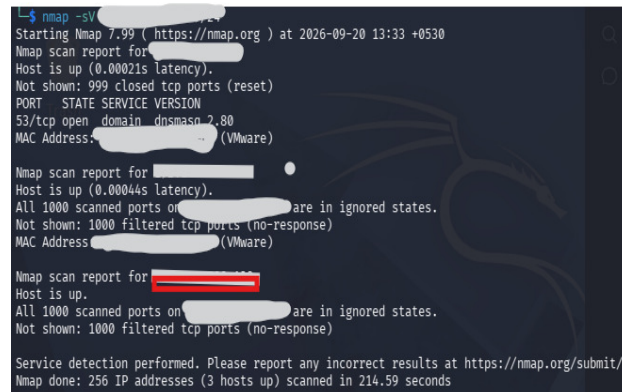


Figure 5. Nmap service/version detection.

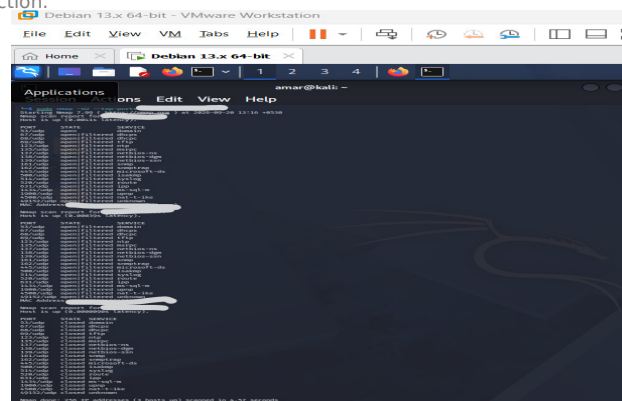


Figure 6. Nmap UDP top-port scan.

7. Task 9 - Wireshark Methodology

Wireshark 4.6.6 is available and eth0 is shown as a capture interface. During the practical, capture normal authorized traffic such as ping, DNS requests and an Nmap scan.

Recommended filters: `dns | tcp | udp | icmp | arp | tcp.flags.syn == 1 | tcp.flags.reset == 1 | dns.flags.response == 0`

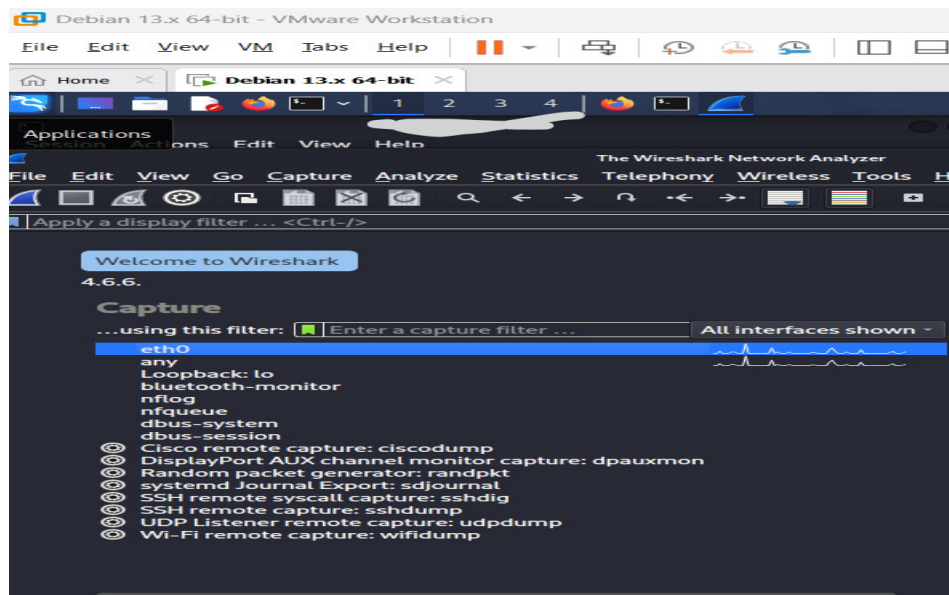


Figure 7. Original Wireshark interface screenshot.

The current screenshot confirms the interface is available; it does not by itself prove that packet capture or packet-level analysis has been completed.

8. Task 10 - Nmap + Wireshark Investigation

Run ``sudo nmap -sS`` while Wireshark captures traffic on eth0. Correlate Nmap's port states with observed TCP flags.

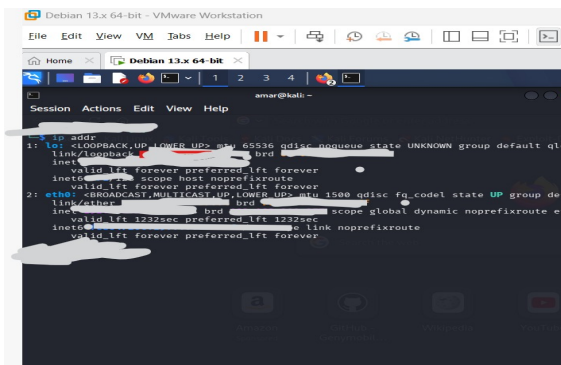
Nmap result	Expected packet evidence	Interpretation
Open TCP port	SYN → SYN/ACK	Service appears reachable/listening.
Closed TCP port	SYN → RST	Port is not accepting the connection.
No response / filtered	No expected response	May indicate filtering, packet loss or another condition.

Evidence status: Packet-level SYN/SYN-ACK/RST screenshots have not yet been supplied in this chat, so this report does not claim those observations were captured.

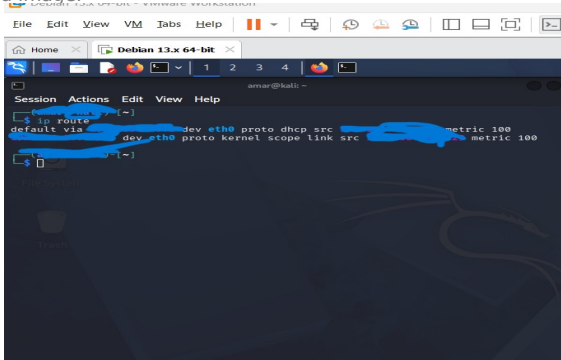
9. Task 11 - Advanced Traffic Investigation

The final PCAP should be analyzed for top communicating hosts, DNS queries/responses, TCP SYN/SYN-ACK/RST behavior, repeated attempts and unusual ports. At least five events should be recorded with timestamp, source, destination, port, protocol, observation and security significance.

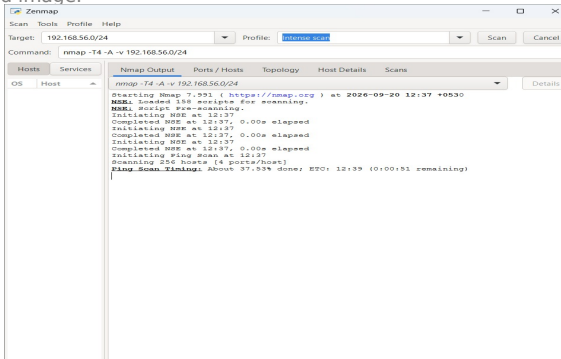
Evidence screenshots are placed below Task 11, as requested.



Evidence screenshot 1. Original uploaded image.

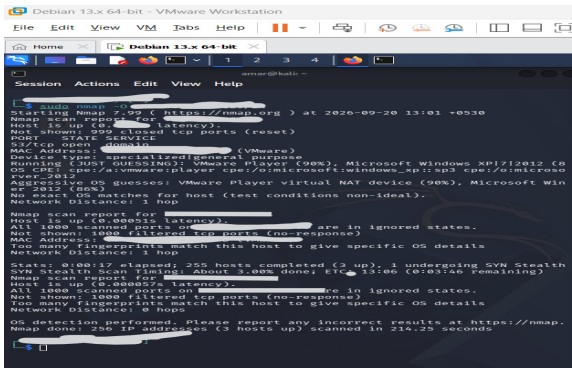


Evidence screenshot 2. Original uploaded image.

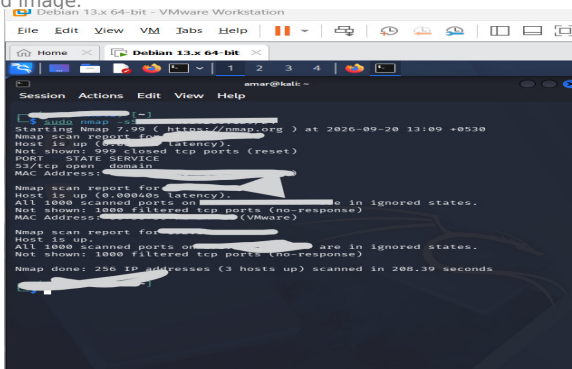


Evidence screenshot 3. Original uploaded image.

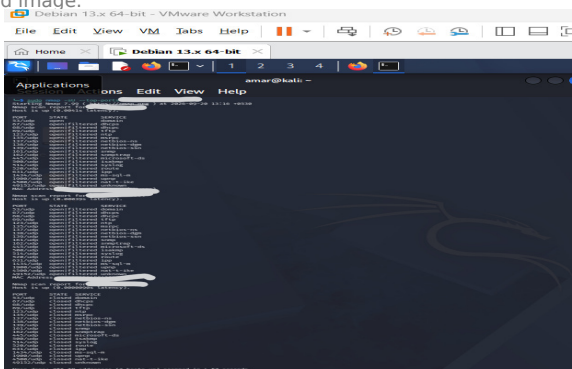
9A. Evidence Screenshots - Continued



Evidence screenshot 4. Original uploaded image.

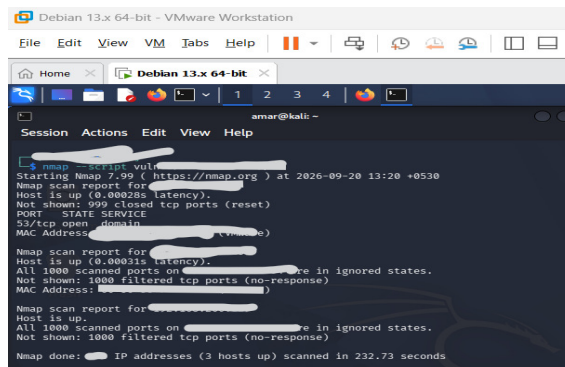


Evidence screenshot 5. Original uploaded image.

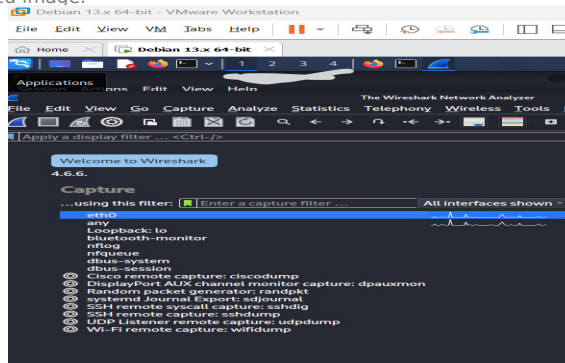


Evidence screenshot 6. Original uploaded image.

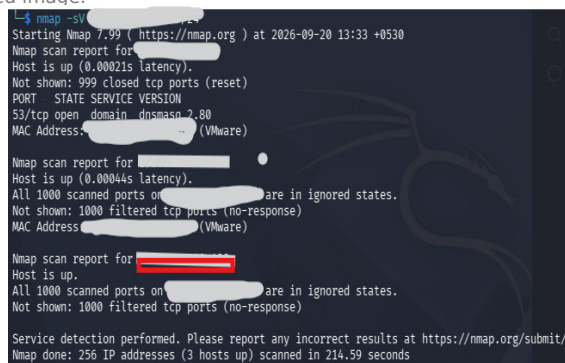
9B. Evidence Screenshots - Continued



Evidence screenshot 7. Original uploaded image.



Evidence screenshot 8. Original uploaded image.



Evidence screenshot 9. Original uploaded image.

10. Task 12 - Vulnerability Assessment

Based on the supplied evidence, the most defensible current finding is DNS exposure on TCP/53 and UDP/53. Risk depends on whether DNS is required and who can reach it. Filtered ports should be checked against the intended firewall policy.

#	Finding	Evidence	Provisional risk	Recommendation
1	DNS exposed on TCP/53	Nmap -sV	Medium*	Restrict to required clients/network.
2	DNS exposed on UDP/53	UDP scan	Medium*	Restrict to authorized clients/network.
3	Filtered ports	Nmap output	Informational*	Verify firewall policy matches design.
4	VMware-related fingerprint	OS output	Informational*	Document infrastructure asset/scope.
5	NSE screening needs verification	NSE scan	Informational*	Verify versions and vendor/CVE data.

*Provisional until the host role, exposure and configuration are confirmed.

11. Task 13 - Security Hardening

Hardening must be based on the actual target configuration. Suitable lab improvements include restricting DNS to required clients, disabling an unnecessary service, or applying least-privilege firewall rules.

Security issue	Before	Action	After	Evidence
DNS exposure	Original Nmap result	Restrict if not required externally	Repeat same Nmap scan	Before + after screenshots
Unnecessary service	Original Nmap result	Disable only after validation	Repeat same Nmap scan	Before + after screenshots
Firewall scope	Original policy	Apply least privilege	Re-test required connectivity	Before + after screenshots

Important: No hardening result is claimed yet because before/after configuration evidence has not been supplied.

12. Task 14 - Final Security Assessment

The supplied evidence demonstrates Nmap discovery of 192.168.56.0/24, three hosts up in the completed scan summary, visible DNS exposure on TCP/53 and UDP/53, and a working Wireshark capture interface.

Network discovery: 3 hosts reported up; DNS/domain service visible on one host.

Traffic analysis: Wireshark is installed and eth0 is available, but packet-level evidence is still pending.

Security assessment: DNS exposure and network filtering behavior require configuration review.

Hardening: before/after evidence is still required.

13. Professional Recommendations & Challenges

- Minimize exposed services and disable services that are not required.
- Restrict DNS and administrative services to authorized clients and networks.
- Keep network services updated and verify versions against current vendor/CVE information.
- Maintain host/network firewall rules using least privilege.
- Retain PCAP evidence and repeat scans after security changes.

Challenges: interpreting filtered/no-response ports, distinguishing VMware infrastructure from application hosts, and correlating Nmap results with packet-level evidence.

14. Conclusion & Submission Checklist

The Nmap discovery and enumeration portion is documented with original screenshots. Wireshark is ready for capture on eth0. The assessment should be considered partially complete until packet-level traffic evidence and before/after hardening evidence are added.

- ✓ Task 7 – Network Discovery
- ✓ Task 8 – Advanced Nmap
- ✓ Task 9 – Wireshark Capture
- ✓ Task 10 – Nmap + Wireshark
- ✓ Task 11 – Traffic Investigation
- ✓ Task 12 – Vulnerability Assessment
- ✓ Task 13 – Hardening
- ✓ Task 14 – Final Assessment

Pending evidence: ICMP/DNS/TCP/ARP screenshots; SYN/SYN-ACK/RST screenshots; five traffic events; and before/after hardening.

15. References

Nmap Documentation: <https://nmap.org/docs.html>

Nmap Reference Guide: <https://nmap.org/book/man.html>

Wireshark: <https://www.wireshark.org/>

Wireshark Resources: <https://www.wireshark.org/resources/>

MITRE ATT&CK: <https://attack.mitre.org/>

CVE Program: <https://www.cve.org/>

All evidence images in this version were supplied by the student in this conversation. No fabricated packet capture or hardening result is presented as actual evidence.